

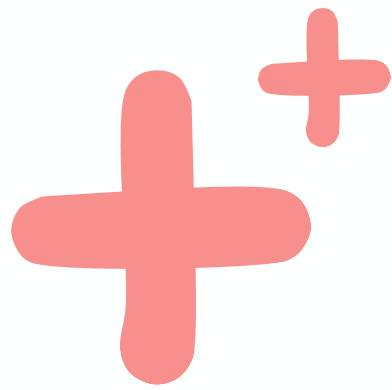


Introduction to Data Science

# PREDICTING CARDIAC DISEASES IN CHILDREN

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# Initial Data Exploration

This step was made to check the integrity of the data and ensure its suitability for predictive modeling by identifying structural issues and quality concerns. The main observations are listed as follows:

**Presence of Impossible Values**

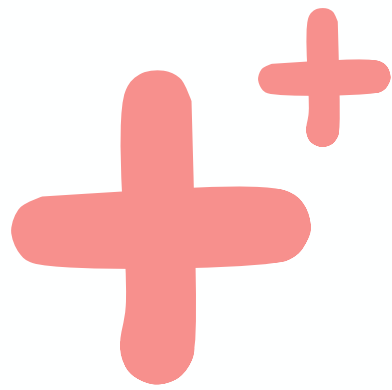
**Incorrect Variable Formatting**

**Significant Missing Data**

**Categorical Noise**

**Imbalanced Target Variable**





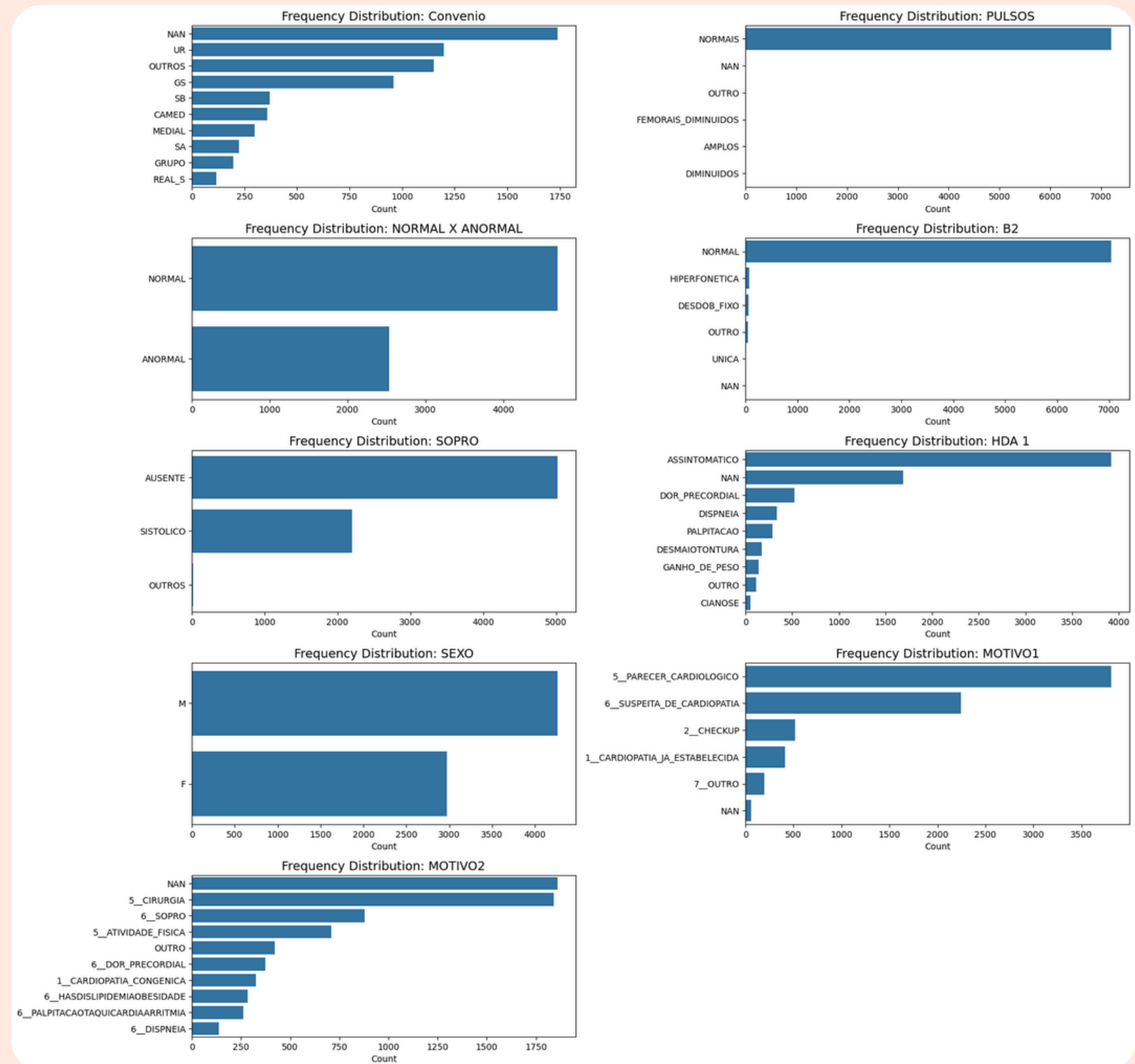
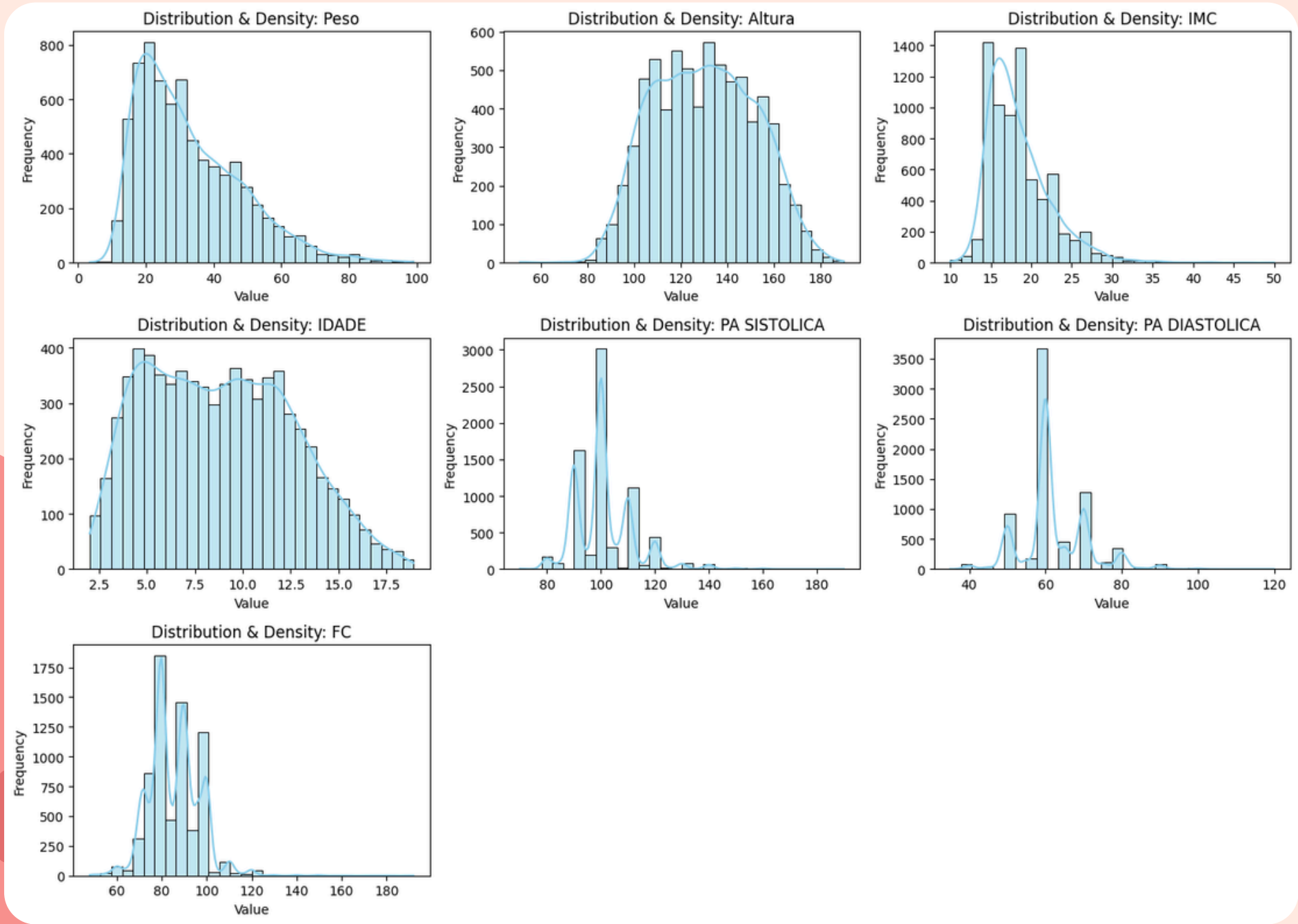
# Data Cleaning

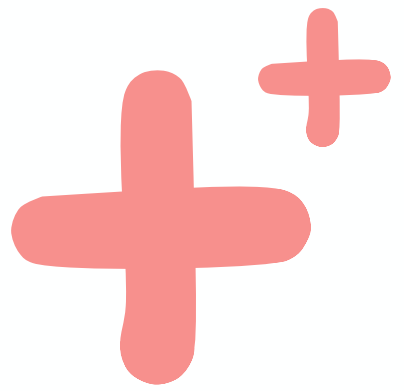
Based on the initial integrity checks, a rigorous cleaning pipeline was applied to resolve inconsistencies and prepare a high-quality dataset for machine learning.

- **Plausibility Filtering:** The dataset was restricted to a pediatric range (ages 2 to 19) and filtered to exclude biologically impossible measurements for weight, height, and heart rate.
- **Label Standardization:** Categorical entries were unified (e.g., consolidating sex labels to 'M' and 'F') and normalized to remove formatting inconsistencies.
- **Data Pruning:** Irrelevant columns such as ID, DN and ATENDIMENTO were removed, along with features showing excessive missingness (like HDA2 and PPA) and duplicate patient records.
- **Clinical Logical Checks:** Records were validated against clinical logic, such as ensuring systolic pressure was not lower than diastolic pressure.
- **Feature Refinement:** Rows with more than 50% missing data were discarded to maintain the robustness of the analytical base.



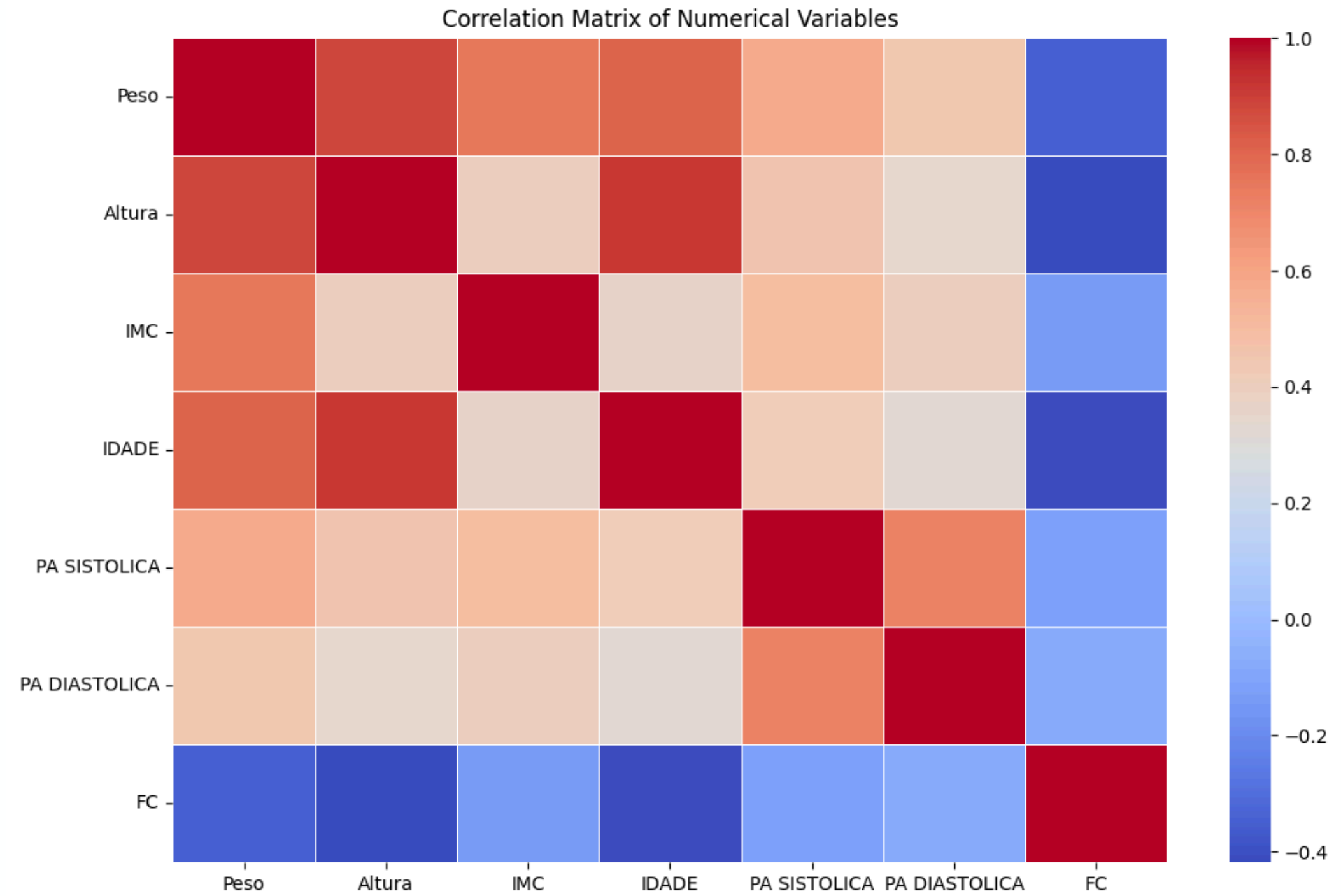
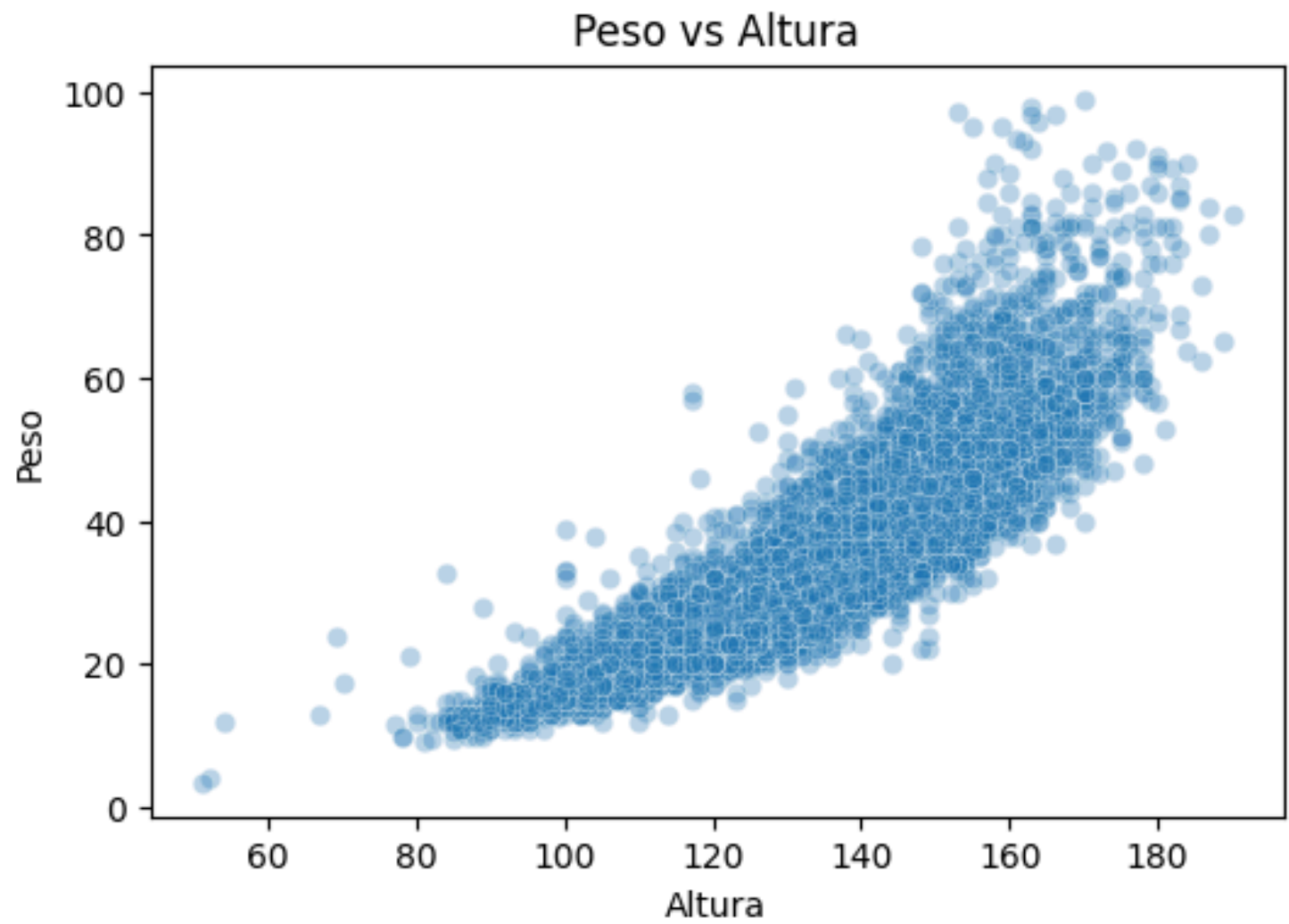
# Univariate Analysis





# Bivariate Analysis

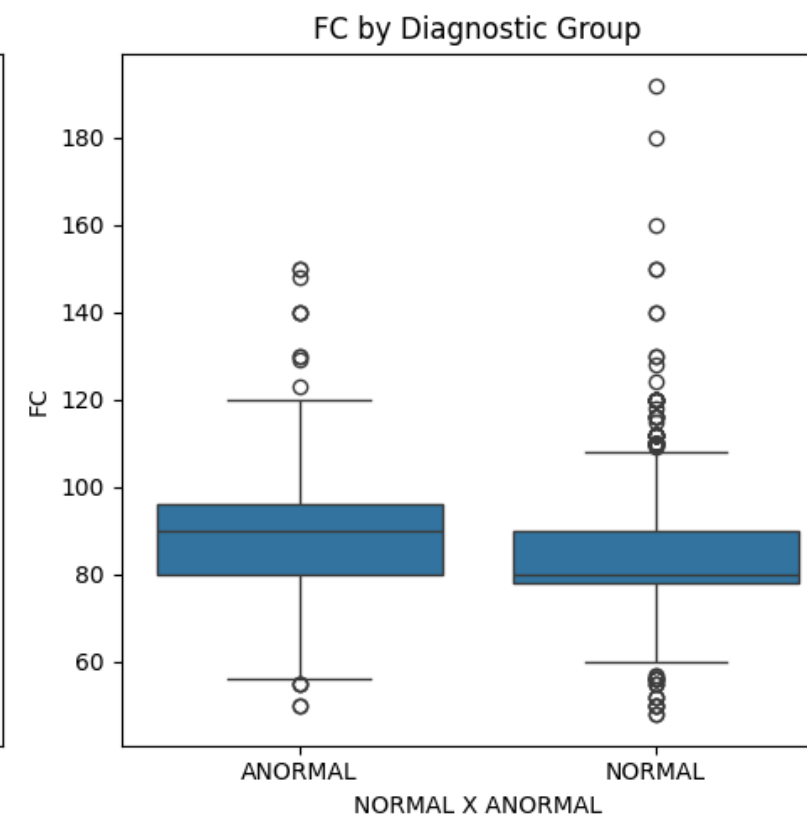
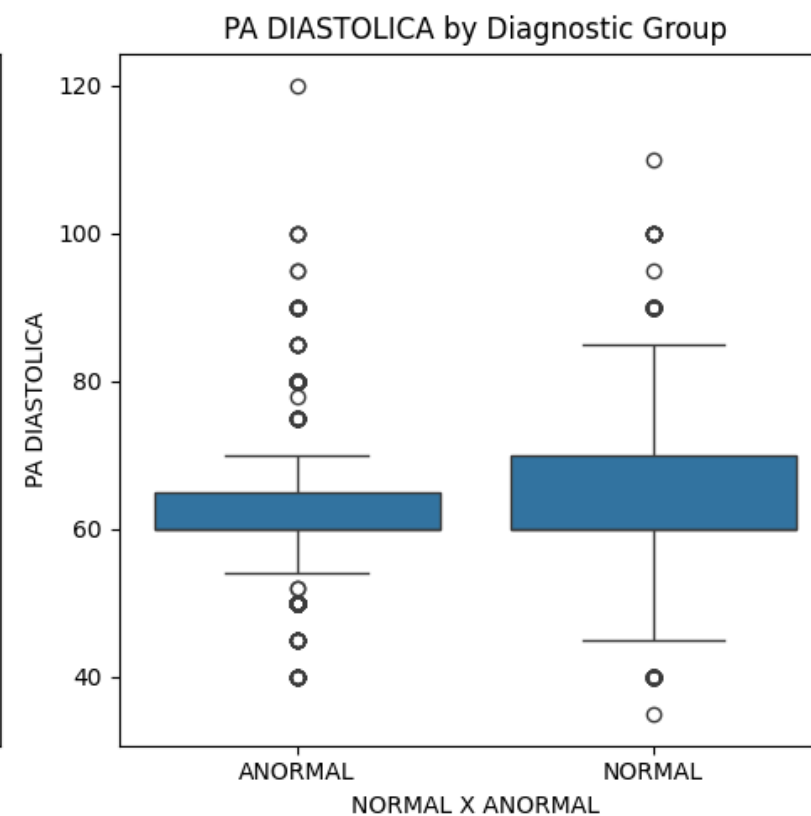
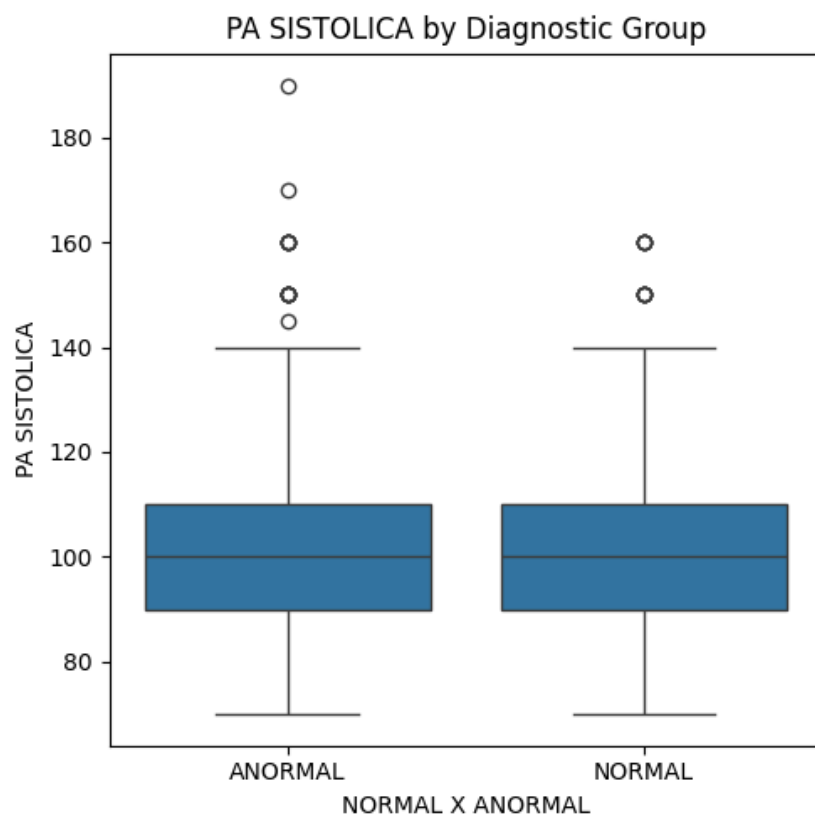
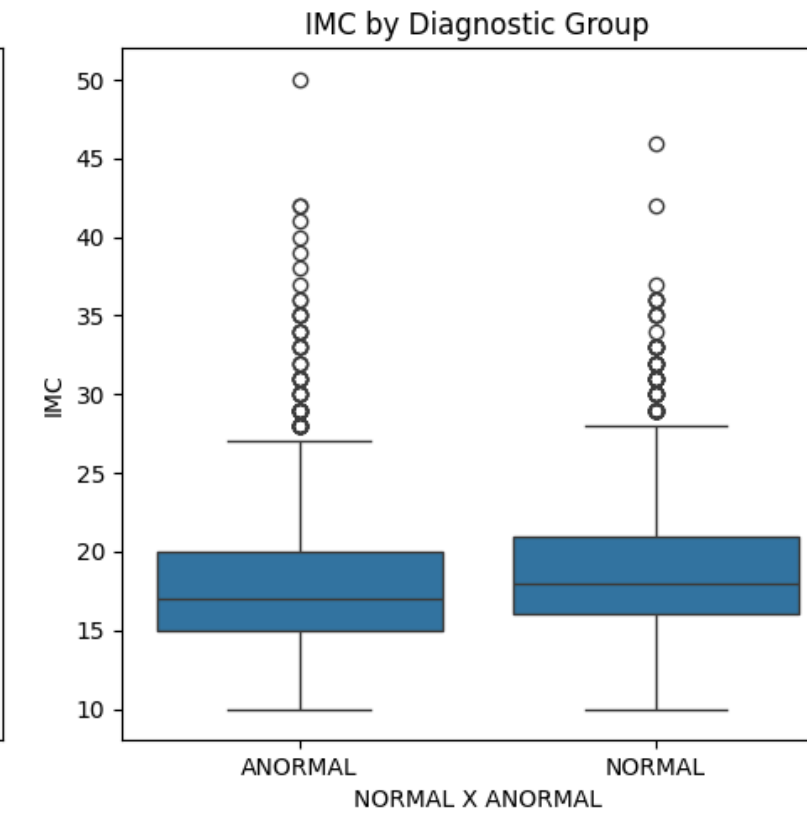
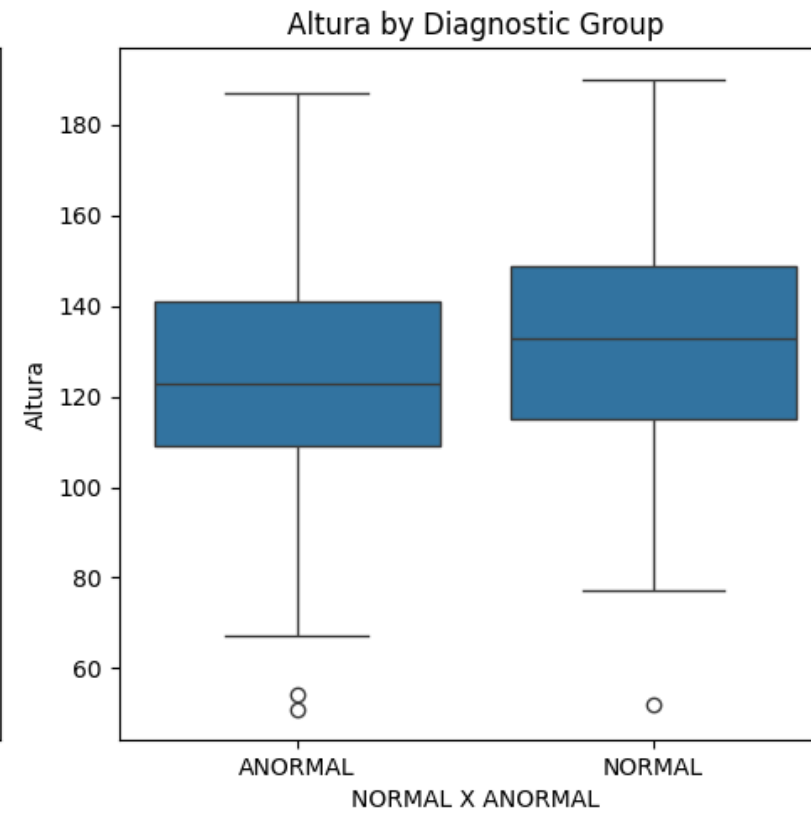
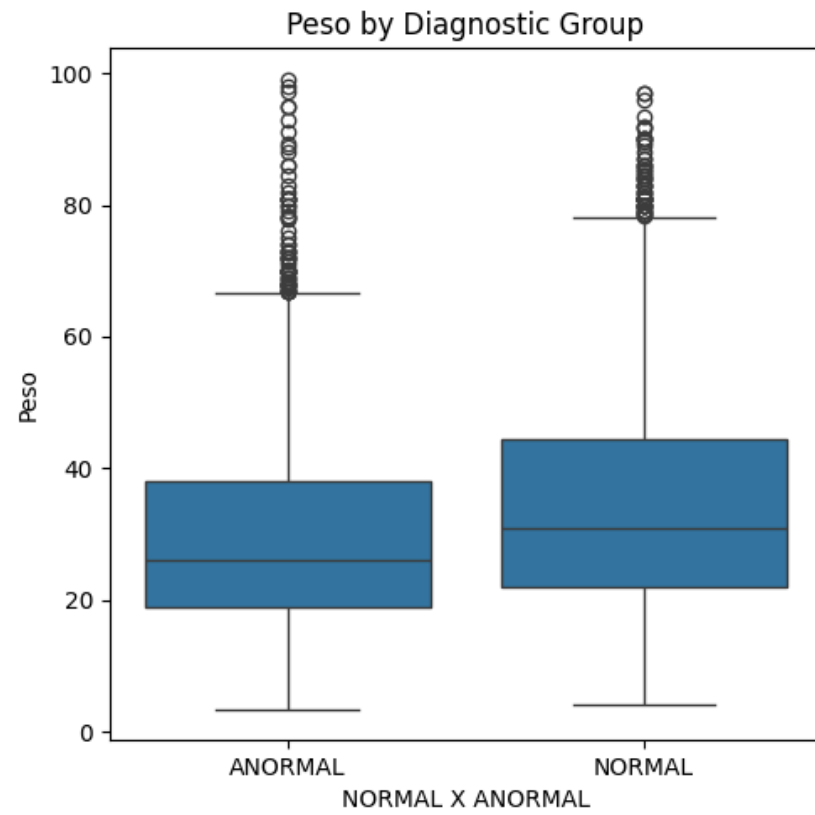
Numeric-numeric (redundancy)

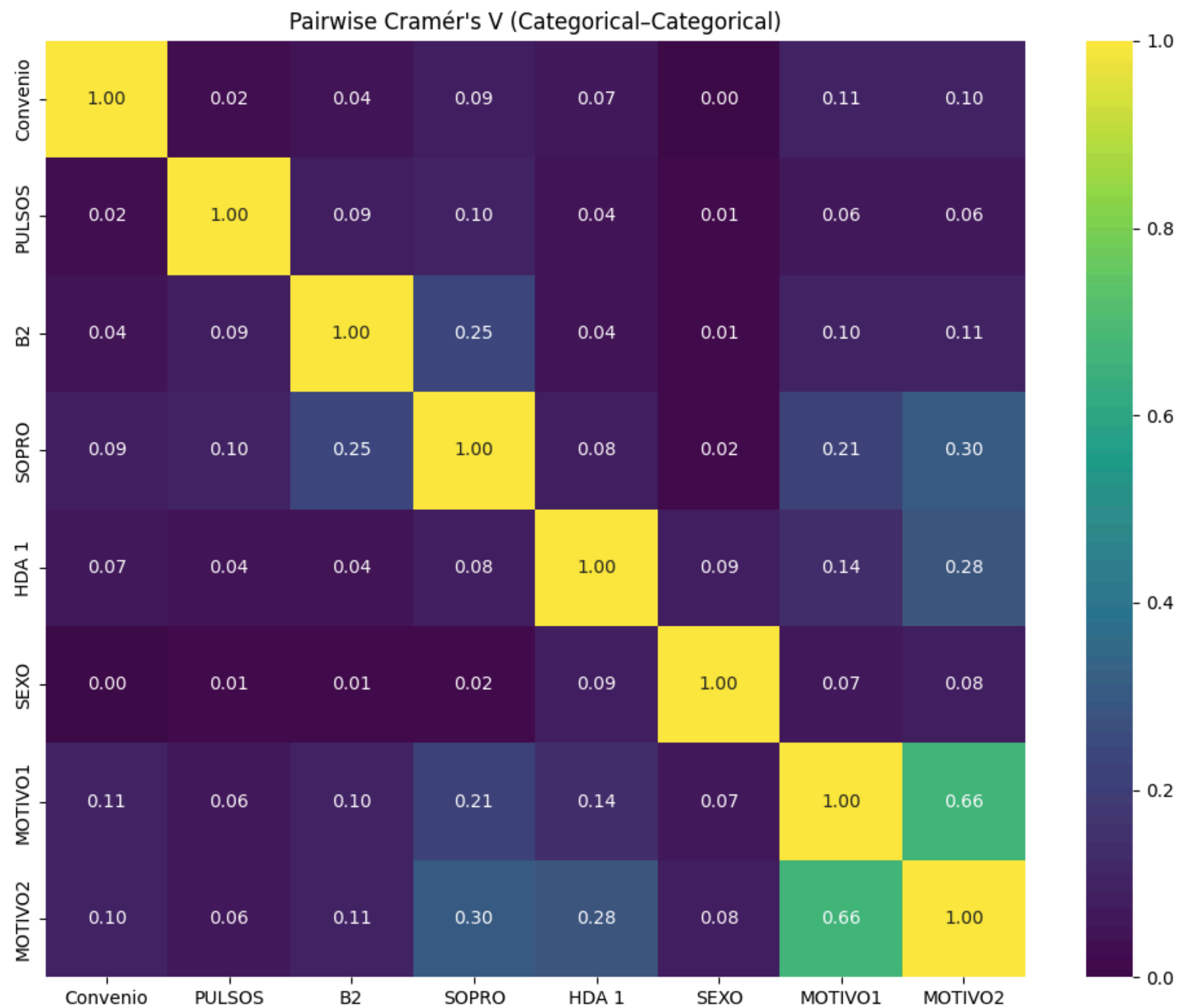
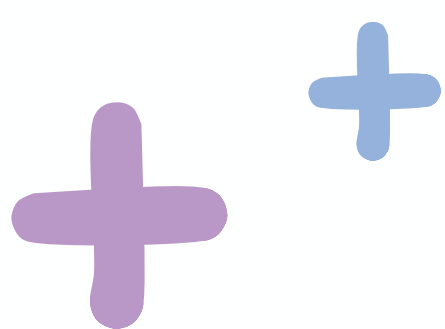




# Bivariate Analysis

Numeric-target





# Bivariate Analyses

Categorical-Categorical

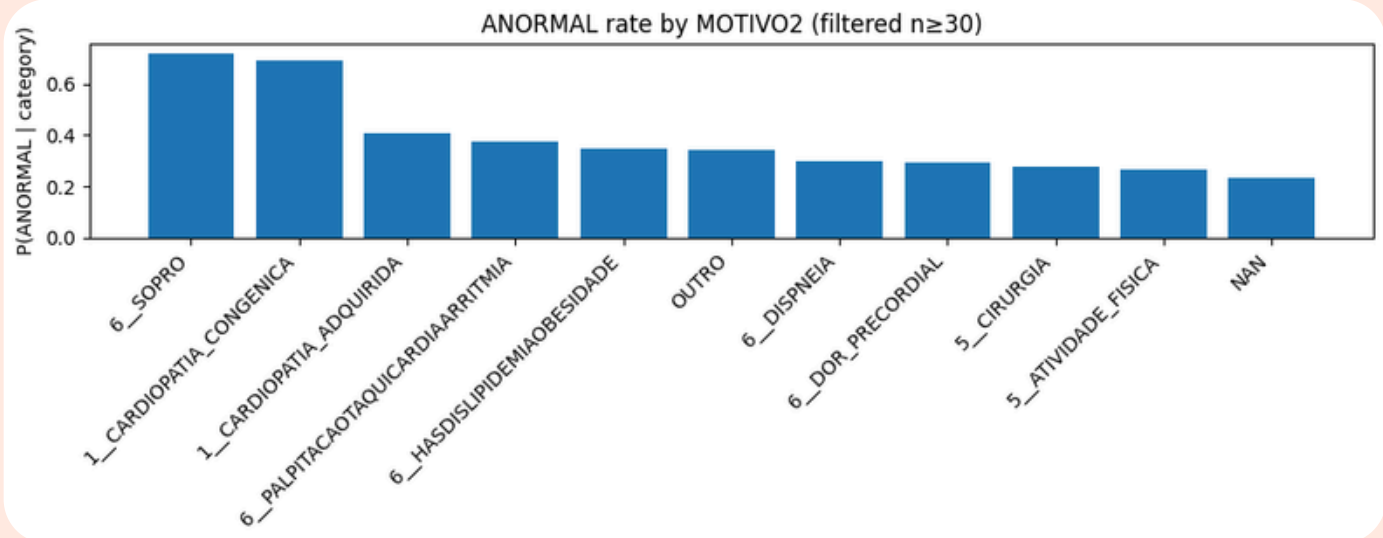
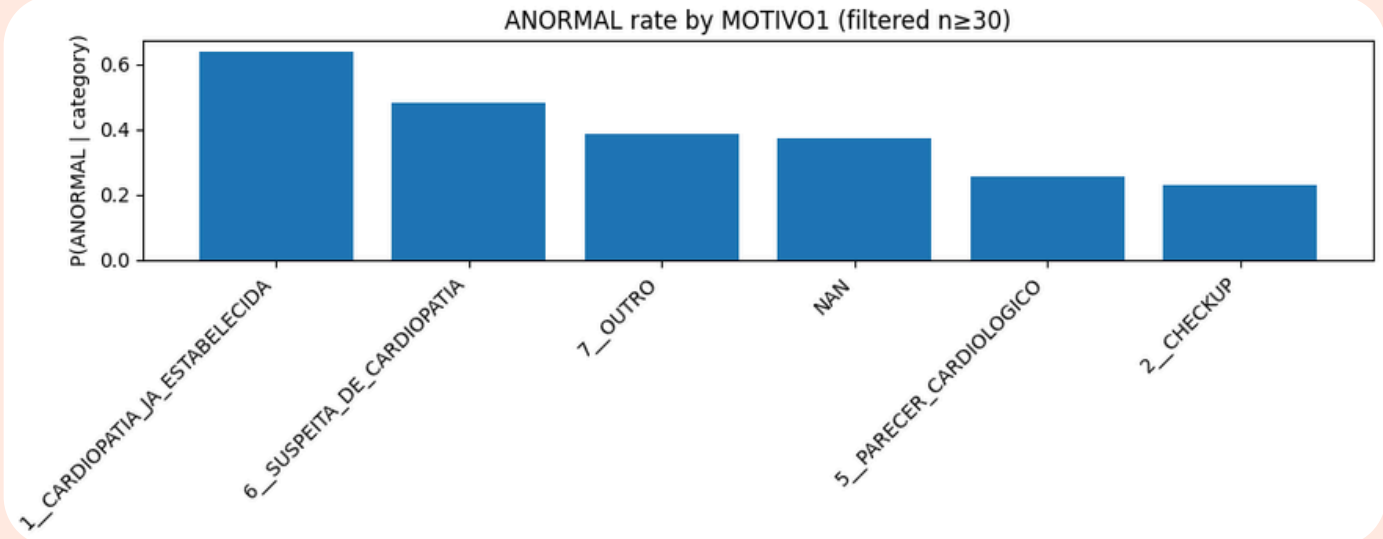
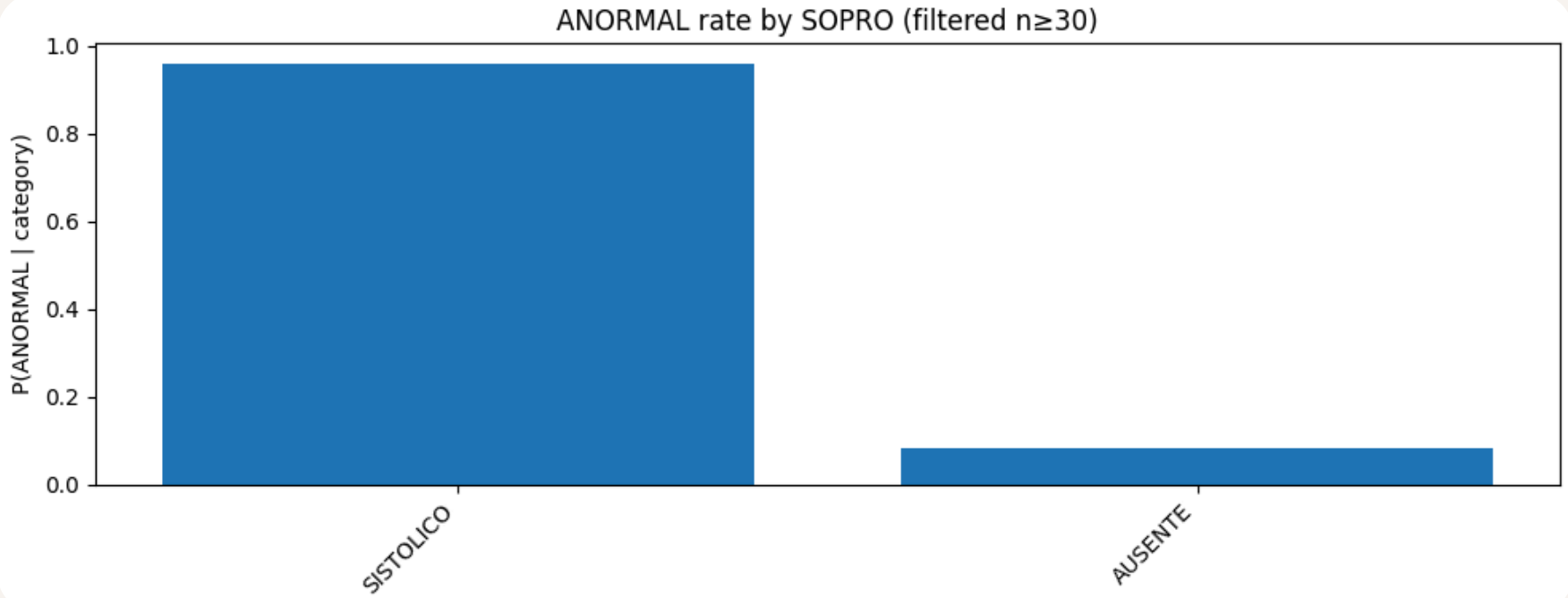




# Bivariate Analysis

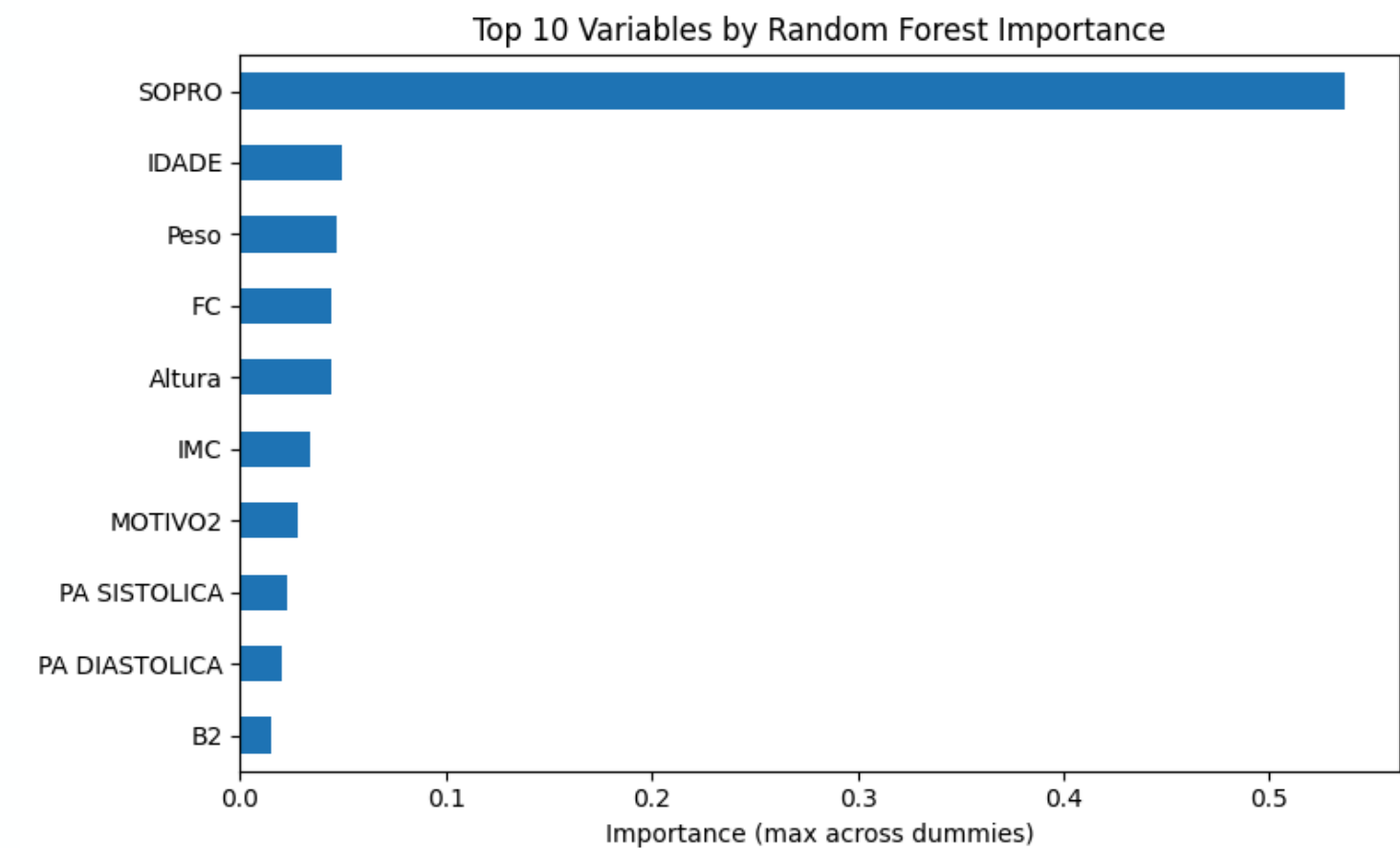
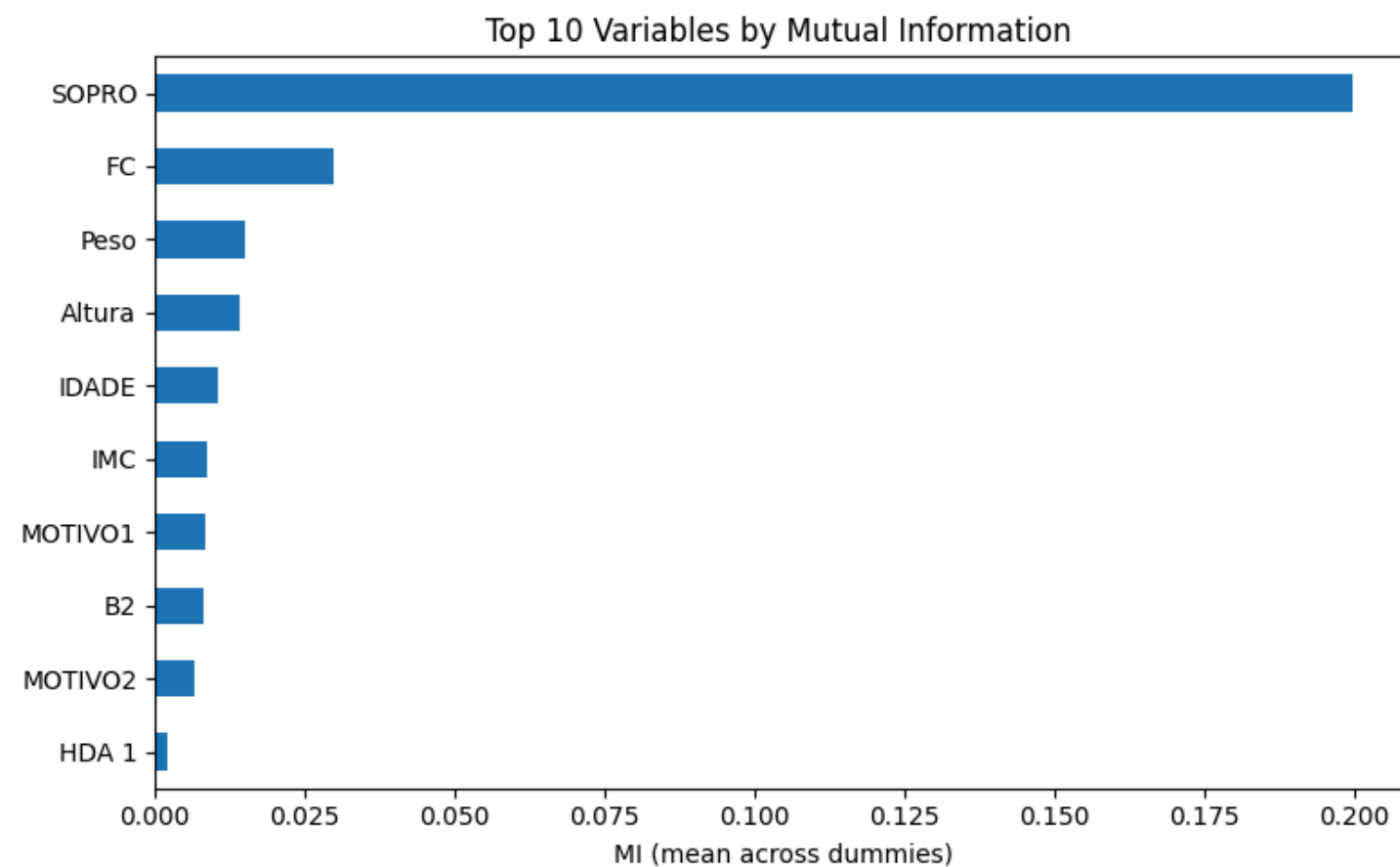
## Categorical-target

|   | Variable | n_levels | CramersV | chi2_p_value  | Top_category_by_ANORMAL_rate    | Top_ANORMAL_rate |
|---|----------|----------|----------|---------------|---------------------------------|------------------|
| 1 | SOPRO    | 3        | 0.844019 | 0.000000e+00  | SISTOLICO                       | 0.957766         |
| 6 | MOTIVO2  | 16       | 0.351379 | 1.090040e-180 | 5_USO_DE_CISAPRIDA              | 1.000000         |
| 5 | MOTIVO1  | 6        | 0.263447 | 2.951991e-106 | 1_CARDIOPATIA_JA_ESTABELECIDADA | 0.639024         |
| 2 | B2       | 6        | 0.183959 | 7.272464e-51  | HIPERFONETICA                   | 0.970149         |
| 4 | Convenio | 20       | 0.139007 | 2.026297e-20  | SUL_AMERICA                     | 0.529412         |
| 7 | HDA 1    | 9        | 0.111137 | 6.334580e-16  | CIANOSE                         | 0.415094         |
| 3 | PULSOS   | 6        | 0.068444 | 2.511057e-06  | DIMINUIDOS                      | 1.000000         |
| 0 | SEXO     | 2        | 0.000000 | 1.000000e+00  | M                               | 0.350857         |

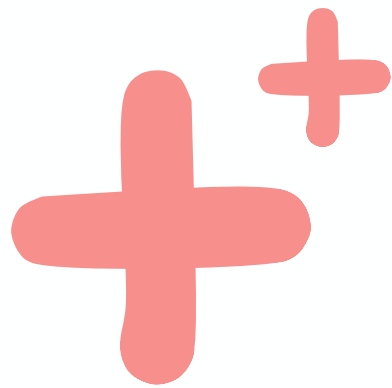




# Multivariate Analysis



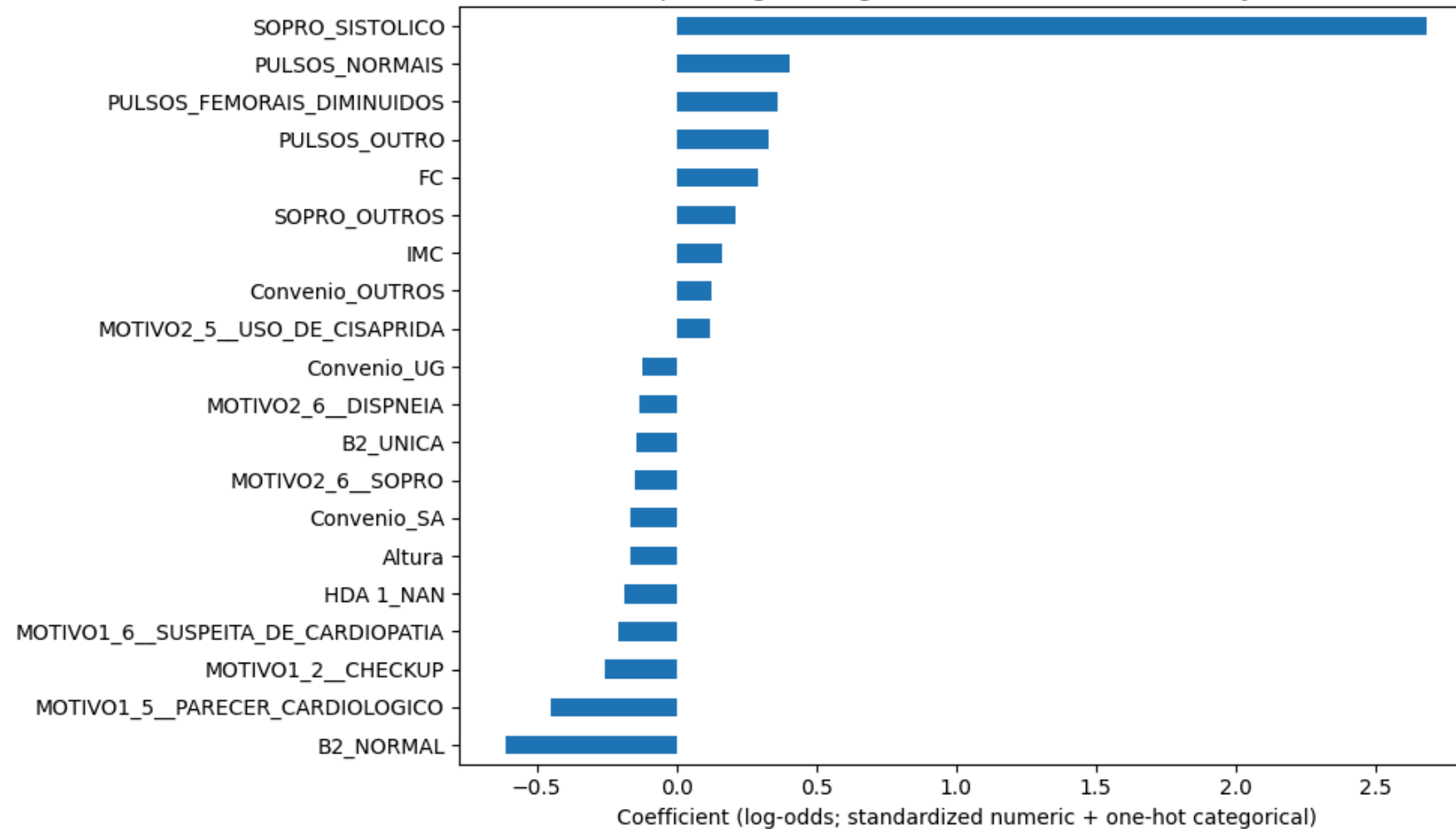
|               | MI_Score | RF_Importance | MI_Scaled | RF_Scaled | Combined_Score |
|---------------|----------|---------------|-----------|-----------|----------------|
| SOPRO         | 0.199661 | 0.536396      | 1.000000  | 1.000000  | 1.000000       |
| FC            | 0.029789 | 0.044495      | 0.149195  | 0.081791  | 0.115493       |
| Peso          | 0.015016 | 0.046739      | 0.075205  | 0.085980  | 0.080593       |
| Altura        | 0.014088 | 0.044416      | 0.070558  | 0.081644  | 0.076101       |
| IDADE         | 0.010430 | 0.049742      | 0.052238  | 0.091587  | 0.071913       |
| IMC           | 0.008802 | 0.034041      | 0.044085  | 0.062277  | 0.053181       |
| MOTIVO2       | 0.006736 | 0.028077      | 0.033737  | 0.051145  | 0.042441       |
| B2            | 0.008068 | 0.014716      | 0.040409  | 0.026204  | 0.033306       |
| MOTIVO1       | 0.008343 | 0.013262      | 0.041787  | 0.023491  | 0.032639       |
| PA SISTOLICA  | 0.000000 | 0.023043      | 0.000000  | 0.041749  | 0.020874       |
| PA DIASTOLICA | 0.000000 | 0.020004      | 0.000000  | 0.036075  | 0.018038       |
| HDA 1         | 0.002141 | 0.012141      | 0.010725  | 0.021399  | 0.016062       |



# Multivariate Analysis

Logistic regression (joint-effect baseline)

Top 20 Logistic Regression Coefficients (train-only fit)



|          | MI       | RF       | REG      | MI_s     | RF_s     | REG_s    | Combined_3way |
|----------|----------|----------|----------|----------|----------|----------|---------------|
| SOPRO    | 0.199661 | 0.536396 | 2.684996 | 1.000000 | 1.000000 | 1.000000 | 1.000000      |
| FC       | 0.029789 | 0.044495 | 0.289300 | 0.149195 | 0.081791 | 0.103521 | 0.111502      |
| B2       | 0.008068 | 0.014716 | 0.610535 | 0.040409 | 0.026204 | 0.223728 | 0.096780      |
| MOTIVO1  | 0.008343 | 0.013262 | 0.452335 | 0.041787 | 0.023491 | 0.164529 | 0.076602      |
| Altura   | 0.014088 | 0.044416 | 0.167804 | 0.070558 | 0.081644 | 0.058056 | 0.070086      |
| Peso     | 0.015016 | 0.046739 | 0.024172 | 0.075205 | 0.085980 | 0.004309 | 0.055165      |
| IMC      | 0.008802 | 0.034041 | 0.163233 | 0.044085 | 0.062277 | 0.056346 | 0.054236      |
| PULSOS   | 0.000949 | 0.000678 | 0.402740 | 0.004751 | 0.000000 | 0.145970 | 0.050241      |
| IDADE    | 0.010430 | 0.049742 | 0.012658 | 0.052238 | 0.091587 | 0.000000 | 0.047942      |
| MOTIVO2  | 0.006736 | 0.028077 | 0.151054 | 0.033737 | 0.051145 | 0.051789 | 0.045557      |
| HDA 1    | 0.002141 | 0.012141 | 0.188864 | 0.010725 | 0.021399 | 0.065937 | 0.032687      |
| Convenio | 0.001970 | 0.007220 | 0.163539 | 0.009868 | 0.012211 | 0.056460 | 0.026180      |



# Multivariate Analysis



|         | n    | abnormal_rate |
|---------|------|---------------|
| Cluster |      |               |
| 0       | 5784 | 0.350622      |
| 1       | 2    | 0.500000      |

|         | Peso      | Altura    | IMC       | IDADE    | PA SISTOLICA | PA DIASTOLICA | FC         |
|---------|-----------|-----------|-----------|----------|--------------|---------------|------------|
| Cluster |           |           |           |          |              |               |            |
| 0       | 33.001798 | 130.39834 | 18.409751 | 8.845325 | 100.97666    | 62.151971     | 85.527361  |
| 1       | 14.000000 | 94.00000  | 16.500000 | 3.195000 | 105.00000    | 70.000000     | 115.000000 |

# Feature Selection

## Optimized Top-8 (Collection II):

**{SOPRO, FC, IDADE, Peso, Altura, B2, MOTIVO1, MOTIVO2}**

This set was chosen because these variables appear repeatedly near the top across the Combined Score and the rank-consensus views, and it keeps the model simpler while preserving the main signal. We also avoid including IMC together with Peso and Altura to reduce redundancy.

We evaluated alternative feature sets, however, because their results were very similar, we retained this one.

# Data Preprocessing/Splitting

**The preprocessing and splitting followed a standard way across the three different data collections to ensure consistency and reliability.**

- Target Encoding: Mapped diagnostic outcomes to a binary format, where "NORMAL" is 0 and "ANORMAL" is 1.
- Categorical Imputation: Handled missing text-based data by filling entries with the most frequent value.
- One-Hot Encoding: Converted categorical variables into numerical dummy variables to allow for mathematical modeling.
- Stratified Splitting: Divided the dataset into training (80%) and testing (20%) sets, ensuring the proportion of cardiac conditions remained consistent across both groups.
- Numerical Imputation: Filled missing numerical values using the median to prevent outliers from skewing the results.
- Feature Scaling: Applied a standard scaler to normalize all features, ensuring that variables with larger ranges did not disproportionately influence the model.



# Modelling

## All Features

| Model               | Accuracy | Precision | Recall   | F1-Score | ROC-AUC  |
|---------------------|----------|-----------|----------|----------|----------|
| Logistic Regression | 0.918452 | 0.949309  | 0.811024 | 0.874735 | 0.918776 |
| Random Forest       | 0.919834 | 0.955814  | 0.809055 | 0.876333 | 0.930143 |
| XGBoost             | 0.914305 | 0.936364  | 0.811024 | 0.869198 | 0.924721 |
| SVM                 | 0.912923 | 0.934091  | 0.809055 | 0.867089 | 0.916971 |
| Mln                 | 0.906704 | 0.906318  | 0.818898 | 0.860393 | 0.917048 |

## Top 8 Features

| Model               | Accuracy | Precision | Recall   | F1-Score | ROC-AUC  |
|---------------------|----------|-----------|----------|----------|----------|
| Logistic Regression | 0.917761 | 0.947126  | 0.811024 | 0.873807 | 0.925274 |
| Random Forest       | 0.915688 | 0.944700  | 0.807087 | 0.870488 | 0.917646 |
| XGBoost             | 0.914997 | 0.940503  | 0.809055 | 0.869841 | 0.915569 |
| SVM                 | 0.917070 | 0.938914  | 0.816929 | 0.873684 | 0.913168 |
| Mln                 | 0.908086 | 0.921348  | 0.807087 | 0.860441 | 0.917151 |

## Without murmur

| Model               | Accuracy | Precision | Recall   | F1-Score | ROC-AUC  |
|---------------------|----------|-----------|----------|----------|----------|
| Logistic Regression | 0.738079 | 0.696049  | 0.450787 | 0.547192 | 0.761587 |
| Random Forest       | 0.747754 | 0.698061  | 0.496063 | 0.579977 | 0.769963 |
| XGBoost             | 0.713891 | 0.615196  | 0.494094 | 0.548035 | 0.740734 |
| SVM                 | 0.736006 | 0.710000  | 0.419291 | 0.527228 | 0.750130 |
| Mln                 | 0.705598 | 0.579457  | 0.588583 | 0.583984 | 0.731115 |

# Conclusion

The analysis confirms that machine learning can effectively support pediatric cardiac screening, provided that clinical context and data quality are maintained.

- **Data Readiness:** Extensive cleaning and range filtering were essential to transform raw clinical records into a robust dataset for predictive modeling.
- **The Power of SOPRO:** Heart murmur (SOPRO) was identified as the single strongest predictor of cardiac abnormalities.
- **Feature Efficiency:** A compact set of Top-8 features achieves an ROC-AUC above 0.91, proving that a small number of clinical indicators can provide high diagnostic value.
- **Critical Dependency:** Removing heart murmur data causes a major performance drop, showing that murmur information is essential for safe screening.
- **The best model depends on clinical use:** high-recall models are preferable for screening, while high-precision models are preferable for confirmation.
- **Decision Support:** These models are designed to act as a decision support tool for clinicians, not as a replacement for professional physical examinations.





**THANK YOU  
FOR YOUR  
ATTENTION**

